Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	11th October 2026
Team ID	ServiceNow - A2
Project Name	Car Rental Request Automation in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Car Rental Request Submission	Employee can open the Car Rental Request catalog item and submit a request with required details.
FR-2	Request Information Capture	Capture requester, travel purpose, pickup/drop-off locations, travel dates/times, car type and additional requirements.
FR-3	Request Validation & RITM Creation	Mandatory variables are validated and a Requested Item (RITM) is created for the submitted request.
FR-4	Approval Management	The request is routed to the appropriate approver and the approval outcome is recorded.
FR-5	Transport Team Assignment	Approved requests are automatically assigned/routed to the Transport team for fulfillment.
FR-6	Vehicle/Rental Availability	Transport team reviews the request and confirms whether a suitable vehicle/rental arrangement is available.
FR-7	Notifications & Status Tracking	Requester receives submission, approval/rejection, assignment and completion notifications and can track the RITM status.
FR-8	Request Completion	Transport team records fulfillment details and closes/completes the request after the rental arrangement is confirmed.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Catalog form should be simple, clear and easy for employees to complete.
NFR-2	Security	Only authorized users should submit, approve, assign or update requests according to ServiceNow roles and access controls.
NFR-3	Reliability	Automation should process requests consistently and preserve request history and audit information.
NFR-4	Performance	Form submission, RITM creation, routing and notifications should complete without unnecessary delay.
NFR-5	Availability	ServiceNow should provide continuous access to the request process subject to platform availability.
NFR-6	Scalability	The solution should support increasing numbers of employees and car rental requests without redesigning the core process.

Solution Requirements – Car Rental Request Automation

The proposed solution is a ServiceNow-based Car Rental Request Automation system. It provides employees with a standardized Service Catalog form, captures travel and vehicle information, creates a Requested Item (RITM), routes the request for approval, assigns approved requests to the Transport team, sends notifications and maintains status tracking.

Functional Requirements:

Non-functional

Requirements:

Project Design Phase-II
Technology Stack (Architecture & Stack)

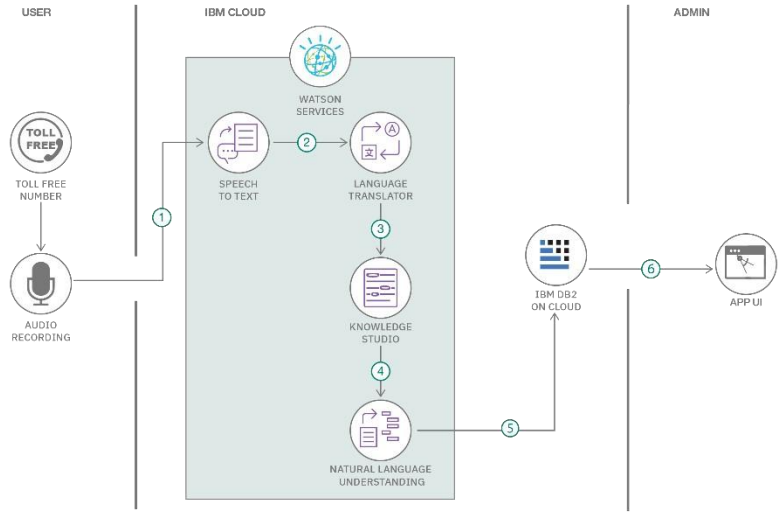
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Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Employee-facing catalog item and request form.	ServiceNow Service Catalog / Catalog Item
2	Request Management	Creates and tracks the service request and Requested Item (RITM).	ServiceNow Request Management
3	Automation Logic	Controls validation, approval, assignment and fulfillment stages.	Flow Designer / Workflow
4	Approval Management	Routes requests to the appropriate approver and records the decision.	ServiceNow Approvals
5	Assignment & Fulfillment	Routes approved requests to the Transport team and records fulfillment information.	Assignment Groups / RITM
6	Notifications	Sends request and status updates to relevant users.	ServiceNow Notifications
7	Data Storage	Stores request, variable, approval, assignment and task information.	ServiceNow Platform Database
8	Access Control	Controls access using roles, ACLs and platform permissions.	ServiceNow Roles / ACLs
9	Deployment	Moves configuration between instances for testing and deployment.	Update Sets

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required for the core solution; ServiceNow platform capabilities are used.	ServiceNow
2	Security Implementations	Role-based access, ACLs, authentication and controlled access to catalog, request and fulfillment records.	ServiceNow Roles / ACLs
3	Scalable Architecture	Supports additional request variables, approval stages and fulfillment rules as demand grows.	ServiceNow Platform
4	Availability	Cloud-hosted ServiceNow platform provides access to the catalog and request records according to platform availability.	ServiceNow
5	Performance	Use mandatory fields, efficient flow conditions and targeted notifications to minimize unnecessary processing.	Flow Designer / ServiceNow

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d

Technical Architecture – Car Rental Request Automation

The architecture uses ServiceNow as the central platform. The employee interacts with the Service Catalog; ServiceNow creates and manages the request/RITM, executes Flow Designer or Workflow automation, routes approvals and assignments, stores request information and sends notifications.

High-Level Architecture Flow

EMPLOYEE / REQUESTER ↓ ServiceNow Service Catalog – Car Rental Request
↓ Catalog Variables → Request / Requested Item (RITM)
↓ Flow Designer / Workflow ➔ Validation → Approval → Assignment
↓ Transport Team * Availability Check → Fulfillment Details → Completion
↓ Notifications & Status Tracking ➔ Requester / Approver / Transport Team

Table-1: Components & Technologies

Table-2: Application Characteristics

References

- ServiceNow Service Catalog documentation
- ServiceNow Flow Designer documentation
- ServiceNow Request / Requested Item (RITM) documentation
- ServiceNow Access Control (ACL) documentation

Project Design Phase-II
Data Flow Diagram & User Stories

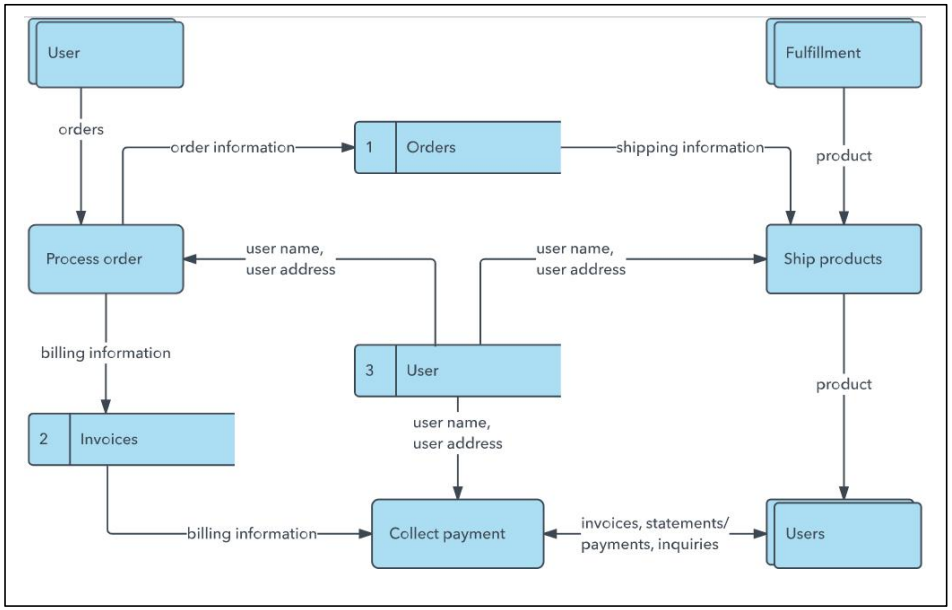
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Data Flow Diagrams:

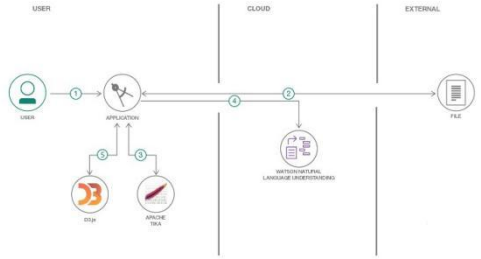
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (Simplified)

Example: DFD Level 0 (Industry Standard)



Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Employee / Requester	Car Rental Request	USN-1	As an employee, I can open the Car Rental Request catalog item so that I can request a vehicle for official travel.	Catalog item is accessible and displays all required request fields.	High	Sprint-1
Employee / Requester	Car Rental Request	USN-2	As an employee, I can enter pickup/drop-off locations, travel dates and times, car type and travel purpose.	Required fields are available and mandatory fields prevent incomplete submission.	High	Sprint-1
Employee / Requester	Car Rental Request	USN-3	As an employee, I can submit the request and receive a request/RITM reference number.	Request is created successfully and RITM number is generated.	High	Sprint-1
Approver	Approval	USN-4	As an approver, I can review the car rental request and approve or reject it.	Approval decision is recorded against the request.	High	Sprint-1
Transport Team	Assignment	USN-5	As a Transport team member, I receive approved car rental requests assigned to my group.	Approved request is automatically routed to the Transport assignment group.	High	Sprint-1
Transport Team	Availability	USN-6	As a Transport team member, I can review the requested dates, locations and car type and confirm availability.	Availability/fulfillment information is recorded on the RITM.	High	Sprint-2
Employee / Requester	Tracking	USN-7	As an employee, I can check the current status of my car rental	Requester can view the current	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
			request.	request/RITM status.		
Employee / Requester	Notifications	USN-8	As an employee, I receive notifications when my request is submitted, approved/rejected and completed.	Relevant notifications are sent at each configured stage.	Medium	Sprint-2
Transport Team	Completion	USN-9	As a Transport team member, I can record fulfillment details and complete the request.	Completion details are saved and the RITM reaches the configured final state.	High	Sprint-2
Administrator	Configuration	USN-10	As a ServiceNow administrator, I can maintain catalog variables, flows, approvals, assignments and notifications.	Configuration can be updated without breaking the request process.	Medium	Sprint-2

Data Flow Diagram – Car Rental Request Automation

The DFD below represents the flow of a car rental request from the employee through ServiceNow approval and Transport team fulfillment. ServiceNow stores the request/RITM and related processing information.

1. Employee / Requester
↓ submits car rental details
2. ServiceNow Service Catalog – Car Rental Request
↓ creates Request / RITM and triggers automation
3. Flow Designer / Workflow
↓ Approval → Assignment → Notification
4. Approver / Transport Team
↓ approve / confirm availability / fulfill
5. Requester receives status updates and final completion notification

User Stories

Project Design Phase-II

Customer Journey Map – Car Rental Request Automation

Date: 11th October 2026
Project Name: Car Rental Request Automation in ServiceNow

This journey map describes the employee experience from identifying the need for a rental car through submission, approval, transport fulfillment and completion.

Stage	Employee Action	Touchpoint	Pain Point	ServiceNow Response	Desired Outcome
1. Need	Employee plans official travel and needs a car.	Travel requirement	Does not know whom to contact or how to request.	Service Catalog provides a single request entry point.	Clear starting point.
2. Submit	Employee enters travel and vehicle details.	Car Rental Request catalog item	Manual forms may miss required details.	Mandatory variables and validation guide the employee.	Complete request submitted.
3. Approval	Employee waits for authorization.	Approval notification	Unclear approval status can cause follow-ups.	Automated approval routing and notifications.	Decision is visible.
4. Assignment	Approved request moves to transport.	RITM / assignment group	Manual forwarding can delay processing.	Approved RITM is routed to Transport team.	Right team receives request.
5. Availability	Transport checks vehicle/rental availability.	RITM fulfillment record	Availability may be uncertain.	Transport records availability and fulfillment details.	Rental arrangement confirmed.
6. Completion	Employee receives final status.	ServiceNow notification/status	Last-minute communication can be difficult.	Completion notification and status tracking.	Employee knows arrangement is complete.